

EK-RA8P1 — Board Specs & What We Use Each Part For

REALITY FILTER. [Verified] = stated on the Renesas EK-RA8P1 kit page or present in this repo's FSP-generated `ra_cfg/fsp_cfg/bsp/bsp_pin_cfg.h` · [Inference] = reasoned, not directly confirmed · [Unverified] = not confirmed this session · [Decision] = our plan, not a hardware fact · [Speculation] = idea only, nothing committed. Sources: Renesas EK-RA8P1 kit page, <https://www.renesas.com/en/design-resources/boards-kits/ek-ra8p1> · this repo's `bsp_pin_cfg.h` (board pin names) · `context.md` (prior verified MCU facts).

1. Silicon specs (MCU: R7KA8P1KFLCAC)

Item	Value	Status
Primary core	Arm Cortex-M85 @ 1 GHz , Helium/MVE SIMD, FPU (single-precision)	[Verified]
Secondary core	Arm Cortex-M33 @ 250 MHz (no Helium, no NPU)	[Verified] core+clock; [Inference] on no-NPU coupling
NPU	Ethos-U55 @ 500 MHz, 256 GOPS , INT8/INT16, inference only (never trains)	[Verified] per <code>context.md</code> ; MAC count 256 [Inference]
On-chip non-volatile	1 MB MRAM (code + model slots)	[Verified]
On-chip SRAM	2 MB SRAM with ECC (activations, buffers)	[Verified]
TCM	256 KB (M85) / 128 KB (M33)	[Verified – <code>context.md</code>]
Process	22ULL (22 nm ultra-low-leakage)	[Verified – <code>context.md</code>]
Security	TrustZone	[Verified – <code>context.md</code>]
Radio	None. No onboard BLE/WiFi on RA8P1	[Verified]

2. Board-level resources (EK-RA8P1 kit)

Board resource	Detail	Status
External Octo-SPI Flash	64 MB (OSPI_* pins, shared with Arduino/Pmod1 via SW4)	[Verified]
External SDRAM	64 MB (32-bit bus, SDRAM_* pins)	[Verified]
Onboard microphones	PDM MEMS mic (MIC_CLK, MIC_DAT)	[Verified]
Audio codec	codec + speaker out; SSI pins AUDIO_MCLK/BCLK/WCLK/SSITX/SSIRX	[Verified]
Ethernet	RGMII + RJ45 (ETHERNET_*, MDIO/MDC, PHY reset)	[Verified]

Board resource	Detail	Status
USB	High-Speed host/device (USB-C) + Full-Speed host/device	[Verified]
Display	2-lane MIPI DSI connector (MIPI_*, DISP_*) + parallel graphics (PARLCD_*); kit ships a 7" 1024×600 capacitive-touch MIPI expansion board	[Verified]
Camera	Camera expansion connector (CAMERA_IRQ, CAM_XCLK, CAMERA_RESET, DCAM_D*/PARCAM_*); kit ships an Arducam OV5640 5 MP module	[Verified]
Expansion headers	Arduino Uno R3 (ARDUINO_AN0..AN5, ARDUINO_D*), 2× Digilent Pmod (SPI/UART/I ² C, PMOD1_*/PMOD2_*), mikroBUS (*_MIKROBUS_*, unpopulated), Qwiic (unpopulated), 2× Grove I ² C + analog (GROVE2_AN0/AN1, unpopulated)	[Verified]
Serial buses	I3C (I3C_SCL/SDA), system I ² C (SYS_I2C_*), SCI UARTs via Pmod/Arduino, VCOM UART to J-Link (VCOMM_*)	[Verified]
Debug	onboard SEGGER J-Link , SWD/JTAG/ETM/SWO	[Verified]
User I/O	3 user LEDs (red/blue/green), 2 user buttons, reset button	[Verified]

Note on unpopulated headers [Verified]: Qwiic, Grove, and mikroBUS are listed as *not populated* on the kit — using them means soldering the connector or going through Pmod/Arduino instead. Several pins are **shared** (OSPI ↔ Arduino/Pmod1, PARLCD ↔ parallel camera) and selected by **SW4**; two subsystems on the same SW4 group cannot both be live. Plan around this.

3. Slot allocation — what we intend to use each part for [Decision]

Nothing below is measured yet; these are assignments, not results.

3.1 Compute

Part	Our use
M85 @ 1 GHz + Helium	Hard-real-time sensor ingest (audio + SpO ₂ + accel), DSP front end (CMSIS-DSP FFT / log-mel), epoch windowing, issues NPU inference, drives the display.
Ethos-U55 NPU	INT8 forward pass of the multimodal apnea model only. Never backprop. Ops Vela can't take fall back to M85 CMSIS-NN.
M33 @ 250 MHz	Background on-device training of the model head (FP32), federation client, store-and-forward, and the egress gate (weights out, raw never).
TrustZone	Deferred: optional hardware-backed egress gate. Only if v1 finishes early. [Decision – defer]

3.2 Memory

Part	Our use
M85 TCM 256 KB	Audio/PPG/accel DMA ring buffers + hottest DSP working set.
M33 TCM 128	Trainable head weights + gradients + Adam state + one micro-batch. Budget ≤ 96 KB; must be

Part	Our use
KB	benchmarked. [Unresolved]
2 MB SRAM (ECC)	NN activations streamed by the NPU + per-modality feature buffers.
1 MB MRAM	Firmware + A/B model slots (atomic OTA swap of the global model; never mutate the live slot).
64 MB SDRAM	Long-horizon working memory: multi-minute multimodal epoch buffers, overnight event/feature history for the AHI tally.
64 MB OSPI Flash	Non-volatile bulk: dataset replay for the demo (PSG recordings played back into the pipeline), event logs, calibration.

3.3 I/O — the multimodal front end

Part	Our use	Status
Onboard PDM MEMS mic (MIC_CLK/MIC_DAT)	Primary contactless modality: snoring / breathing-sound stream, log-mel features.	[Decision]
Audio codec / SSI	Alternative analog audio path + any audible cue or alarm out.	[Decision]
I²C (Pmod / Grove / Qwiic / I3C)	SpO₂ + PPG pulse-oximeter module (desaturation is the classic apnea marker). Exact part not chosen.	[Unverified – part TBD]
I²C or SPI Pmod	Accelerometer for body position + respiratory effort (chest/abdomen motion).	[Decision], part TBD
Arduino analog pins AN0 . . AN5 (ADC)	Analog front-end path if we add ECG or nasal-airflow/thermistor/belt input. Not designed yet.	[Speculation]
Ethernet RGMII	Federation link only : FP32 weights out, global model in, during sync windows. PHY powered down otherwise.	[Decision]
MIPI display (7" touch)	Clinician view: live respiration/SpO ₂ trace, flagged events, running AHI estimate, session start/stop.	[Decision]
Camera connector / OV5640	Contactless chest-motion or IR video modality. Out of v1 scope — privacy cost is high and it competes with PARLCD pins on SW4.	[Speculation]
USB HS	Bulk export of anonymized session summaries / benchmark dumps to the dev PC.	[Decision]
VCOM UART + J-Link/RTT	Telemetry, deadline-miss counters, on-target debug.	[Decision]
User LEDs / buttons	LED = recording/liveness/fault state; buttons = start/stop night session.	[Decision]

3.4 Pin-sharing conflicts to resolve before wiring [Verified from bsp_pin_cfg.h]

1. **OSPI Flash** ↔ **Arduino SPI** ↔ **Pmod1** share pins through **SW4** — if we replay datasets from OSPI, the Arduino SPI slot and part of Pmod1 are unavailable.
2. **Parallel LCD** ↔ **parallel camera** share PARLCD_D*/PARCAM_* — using the parallel camera costs the parallel display path (the MIPI display connector is the way out).

3. **Audio SSI RX** ↔ **DCAM_D2** share P406 (J41) — audio-in and digital camera cannot both claim it.
4. **I3C pull-ups** need **I3C_SEL_L** enabled before the I3C sensor bus works.

4. Federated learning for sleep apnea — the paragraph [Decision]

Sleep apnea is a good federated problem for the same two reasons FoG was, only stronger: overnight polysomnography data is heavily protected patient data that legally cannot be pooled across hospitals (HIPAA/GDPR), and scoring practice, patient mix, sensor placement, and hardware differ enough between sleep labs that a model trained at one site degrades at another [Unverified – must be measured in our own cross-site test, not assumed]. In FedTinyRT each clinic's board is one federation node: it screens its own patients overnight with the current global model, fine-tunes **only the classifier head** on its locally scored epochs on the M33 while the M85/NPU keeps screening in real time, and at the end of a round emits a single **Aggregation_Payload** — dense **FP32 weights of the shared layers, KB-scale** — over Ethernet. A star aggregator runs **FedAvg** across sites, sanity-filters payloads (NaN/Inf, absurd norm, wrong round id), and redistributes **Global_Model[r]**; each board then keeps a **personalized** copy (global + local fine-tune) that is used for local screening and **never submitted**. Raw audio, SpO₂ traces, and clinician labels never cross the federation network. For the contest build, "multiple hospitals" are **subject-level splits of public PSG datasets** — e.g. SHHS, MESA, Apnea-ECG, and the St. Vincent's/UCD database on PhysioNet [Verified – these databases exist and are multimodal/PSG; our specific splits and preprocessing are not yet defined] — replayed from OSPI flash, with one held-out site for a leave-one-hospital-out test. The headline metric is **AUPRC / average precision plus sensitivity & specificity per epoch, and AHI error per night; accuracy is banned** as a headline because apneic epochs are rare. [Decision]

5. Multimodal data on this board — the paragraph [Decision]

The board's value over a single-signal screener is that the EK-RA8P1 gives us **three usable sensing paths at once, natively**: the **onboard PDM MEMS microphone** (snoring, breath sounds, gasp/arousal onsets), an **I²C SpO₂/PPG module** on the Pmod/Grove/Qwiic side (oxygen desaturation and pulse-rate surges, the events an apnea is scored by), and an **I²C/SPI accelerometer** (body position, and chest or abdominal respiratory effort — the signal that separates obstructive from central events). These arrive at wildly different rates — audio at kHz, accel around 100 Hz, SpO₂ around 1 Hz [Inference – typical rates; our exact sampling config is not fixed] — so the M85 timestamps each stream into its own DMA ring buffer in TCM and resamples/aligns them onto a **common epoch grid** (30 s or 60 s epochs, with overlap, matching how apnea is scored clinically [Unverified – final epoch length is a benchmark decision]). Fusion is **per-modality encoder branches into a late-fusion head**: a small 1D-CNN over log-mel audio frames, a light temporal branch over SpO₂/PPG, and a light branch over accel, concatenated into the shared dense layers — so the **frozen convolutional bodies stay on the NPU in INT8** (cheap, and they can be grown for accuracy) while the **small fused head is the only thing that trains on the M33 and the only thing that federates**. Because a sensor can fall off or disconnect mid-night, the design requires **modality-dropout training** (train with branches randomly zeroed) so a missing stream degrades the score instead of breaking inference; if a stream goes dead, the system flags **liveness=false** and **withholds** a prediction rather than emit a false "no event." [Decision] Longer-term/optional modalities — the camera expansion board for contactless chest motion, or an ECG/airflow analog front end on the Arduino ADC pins — are noted in §3.3 as [Speculation], not part of v1.

6. Open items before this page can be signed off

#	Item
1	SpO ₂ /PPG and accelerometer part numbers not chosen; no bench validation of either. [Unverified]
2	Epoch length, overlap, and per-modality sampling rates not fixed. [Unresolved]
3	M33 head-training memory + epoch time not benchmarked (fallback = M85 idle-slot training). [Unresolved]
4	Dataset choice, subject-level "hospital" partitions, and label harmonization across PSG databases not defined.

Item

[Unresolved]

- 5 INT8 vs INT16 for the audio branch, decided by AUPRC, not pre-committed. [Unresolved]
- 6 Fused model dimensions → exact Aggregation_Payload byte size unknown. [Unresolved]
- 7 context.md / RFC-001 / SYSTEM_OVERVIEW.md still describe the FoG target and must be rewritten for apnea.
[Verified – repo state]